

Reach Extension of WDM DP-16QAM Signals Based on Decision-Feedforward Digital Backpropagation

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Abstract We experimentally demonstrate an improved decision-feedforward digital backpropagation (DFF-DBP) to mitigate self- and cross-phase modulation in long-distance WDM systems without using neighboring-channel amplitudes. DFF-DBP with optimized design parameters extends the transmission reach of a 21-channel, 64-Gbaud DP-16QAM system from 2140 km to 2634 km. ©2026 The Author(s)

Introduction

Nonlinear waveform distortion of optical signals in long-distance wavelength-division multiplexed (WDM) transmission systems is a serious problem that limits system performance [1]. Digital backpropagation (DBP) based on the split-step Fourier method (SSFM) is recognized as a promising digital signal processing (DSP) technique to compensate for such distortions. DBP in its original form [2, 3] considers self-phase modulation (SPM) but neglects cross-phase modulation (XPM), and has therefore been extended to address XPM, which can be classified into two groups. The first group [4–8] utilizes the amplitudes of neighboring channels to compensate for the phase shift induced by XPM, denoted by φ_{XPM} hereafter. Although this approach can achieve superior compensation performance, it tends to involve a complicated structure and high computational cost. The other group [9–12] uses only the amplitude of the channel of interest; most of these methods [9–11] perform compensation of φ_{XPM} only once by exploiting its stochastic property. Meanwhile, careful consideration of XPM [13] suggests that φ_{XPM} at a certain position of the transmission line is detectable in the time domain only when chromatic dispersion (CD) has not yet accumulated. As a result, φ_{XPM} must be sequentially compensated immediately after it is generated, as is the case for SPM. To address this requirement, we previously proposed decision-feedforward DBP (DFF-DBP) [12]. DFF-DBP can effectively mitigate nonlinear degradation in WDM signals, particularly those induced by φ_{XPM} , and its operation was validated in a proof-of-concept experiment [12]. However, the experimental conditions in [12] were highly limited, and the potential for reach extension in broadband WDM systems, under optimized launch power conditions and with the application of soft-decision forward error correction (SD-FEC), has not yet been clarified.

In this paper, we experimentally verify that the transmission reach of a 21-channel, 64-Gbaud dual-polarization (DP) 16QAM WDM system can be extended to 2634 km using an improved DFF-DBP design with a four-step-per-span configura-

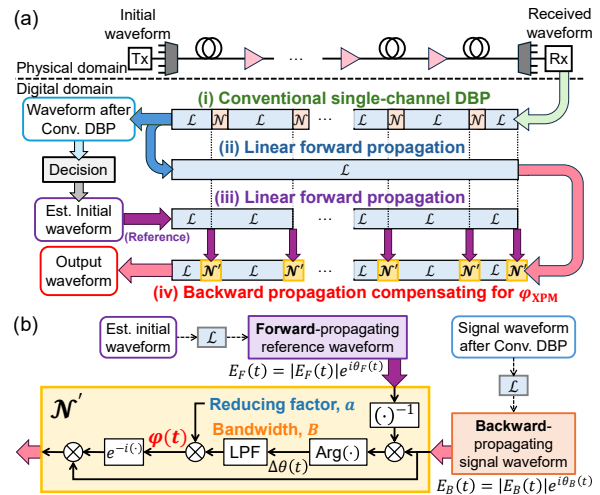


Fig. 1: (a) Schematic of the proposed improved DFF-DBP. $\mathcal{L}$ and $\mathcal{N}$: linear and nonlinear steps of SSFM; $\mathcal{N}'$: nonlinear step accounting for XPM-induced phase shift. (b) Procedure to estimate and compensate for φ_{XPM} .

tion, compared with 2140 km achieved by applying only chromatic dispersion compensation (CDC). To the best of our knowledge, this is the first demonstration of reach extension for WDM DP-16QAM systems comprising more than 20 channels using a DBP-based approach that does not require the amplitudes of neighboring channels.

Improved design of DFF-DBP

Fig. 1(a) depicts a schematic of the proposed improved DFF-DBP architecture. After receiving the signal in the physical domain, four stages of signal processing are performed in the digital domain: (i) Conventional DBP is first applied to compensate for waveform distortion due to SPM. Assuming that the nonlinear distortions caused by SPM and XPM can be treated independently, we focus on XPM hereafter. (ii) The total accumulated CD of the transmission line is reimposed onto the waveform after the conventional DBP. This waveform serves as the input to the final processing stage. The computational cost of this stage is low, since it requires only one linear step of SSFM. (iii) Hard decision is applied to the waveform after the conventional DBP in (i), and an estimate of the initial waveform is reconstructed from the decided symbols. We

then perform linear forward propagation of this estimated initial waveform using the same linear steps as those employed in the conventional DBP in (i). The waveforms after the linear steps serve as reference signals for phase comparison in the next stage. The computational cost of this stage is approximately half that of conventional DBP, since only a single FFT is required to calculate the spectrum. (iv) Backward propagation is then applied to the waveform after (ii), where linear steps in SSFM and phase estimation/compensation steps for φ_{XPM} are alternately repeated. The computational cost of this stage is comparable to that of conventional DBP because the nonlinear step of SSFM for φ_{SPM} is replaced by a similar step for φ_{XPM} . As a result, the overall computational cost of the process shown in Fig. 1(a) is approximately 2.5 times that of conventional DBP, which is reduced compared with the original DFF-DBP [12], which requires approximately three times the cost of conventional DBP.

A step to estimate and compensate for the XPM-induced phase shift φ_{XPM} is shown in Fig. 1(b), which is similar to that of the original DFF-DBP in [12]. Using the backward-propagating signal waveform, $E_B(t) = |E_B(t)| \exp[i\theta_B(t)]$, and the corresponding forward-propagating reference waveform, $E_F(t) = |E_F(t)| \exp[i\theta_F(t)]$, we calculate the phase difference, $\Delta\theta(t) = \theta_B(t) - \theta_F(t)$, followed by low-pass filtering (LPF) with a Gaussian response and a 3-dB bandwidth B to extract the narrowband component of $\Delta\theta(t)$, because $\varphi_{\text{XPM}}(t)$ has a narrow bandwidth when the amount of accumulated CD remains small [13]. We then apply a reducing factor, $0 \leq \alpha \leq 1$, to mitigate the impact of phase cancellation errors caused by decision errors. Finally, we obtain the desired phase shift waveform, $\varphi(t)$, and remove it from $E_B(t)$.

The reducing factor α and bandwidth B of the LPF are key design parameters of DFF-DBP, and they control the trade-off between potential performance gain and the risk associated with using decision-estimated reference signals in calculating $\varphi(t)$. We optimize these parameters based on experimentally obtained received signals to achieve the best overall performance of DFF-DBP.

Experiment

We conducted a recirculation-loop transmission experiment to optimize the design parameters of DFF-DBP and to experimentally verify transmission-reach extension for a 21-channel, 64-Gbaud DP-16QAM WDM system, using the setup shown in Fig. 2. The central channel was defined as the channel under test (CUT) at a center frequency of 193.1 THz, while the other 20 channels were used as dummy signals to induce waveform distortion of the CUT through XPM. The dummy signals had an irregular frequency spacing, as

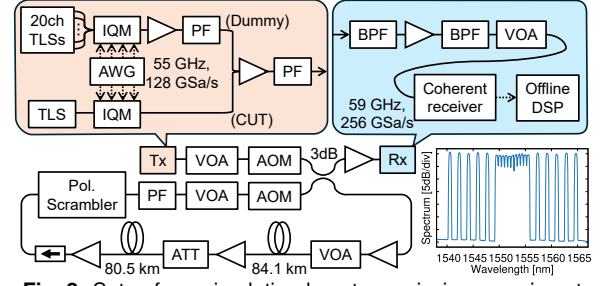


Fig. 2: Setup for recirculation-loop transmission experiment and optical spectrum of 21-channel WDM signal after Tx.

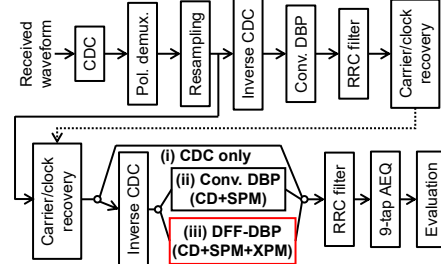


Fig. 3: Procedure for offline DSP including demodulation and three schemes to address transmission impairments.

shown in Fig. 2, where the spacing of the 10 channels adjacent to the CUT was 75 GHz, while that of the remaining 10 channels was 225 GHz. The total bandwidth of the signal was 3 THz, corresponding to 41 channels assuming a regular 75-GHz spacing. This irregular channel arrangement allowed XPM contributions from widely separated channels (> 1 THz) to be included while reducing the total number of channels, thereby relaxing the required maximum output power of the optical amplifiers in the transmission line.

In the transmitter (Tx), the signals were modulated by IQ modulators (IQMs) driven by four-channel electrical waveforms generated by an arbitrary waveform generator (AWG). The generated signal had a DP-16QAM format and was pulse-shaped using a root-raised-cosine (RRC) filter with a roll-off factor of 0.1. The WDM signal was then launched into the recirculation loop. The transmission line consisted of two spans of standard single-mode fiber (SSMF), with lengths of 84.1 and 80.5 km. A polarization scrambler randomly rotated the signal's polarization, and a programmable filter (PF) equalized the optical powers of the channels. The transmission distance of a loop was 164.6 km, and the number of loop circulations was varied from 4 (658 km in total) to 16 (2634 km). The power of the WDM signal launched into the SSMF of each span was varied from -2 to $+5$ dBm/channel. In the receiver (Rx), the CUT was extracted using a band-pass filter (BPF) and detected by a coherent receiver.

The received waveform of the CUT was demodulated using offline DSP, as shown in Fig. 3. In the upper part of Fig. 3, the received waveform was processed in a conventional manner, where CDC, polarization demultiplexing [14], resampling

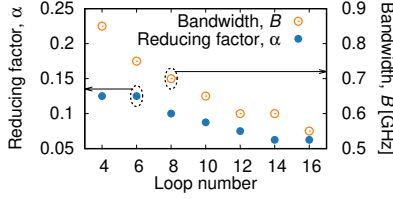


Fig. 4: Optimized parameters of DFF-DBP for the 4-sps configuration as a function of loop number.

to two samples per symbol, conventional DBP with a four-step-per-span (sps) configuration, an RRC filter, and clock and carrier recovery [15] were performed to obtain reliable estimates of the clock and carrier phase. In the lower part, the estimated clock and carrier phase information was applied to the waveform after resampling so that it could serve as the input to the signal processing schemes under study. We then compared three schemes for compensating transmission impairments: (i) CDC only, (ii) conventional DBP, and (iii) DFF-DBP. After an RRC filter and a 9-tap adaptive equalizer (AEQ) to mitigate residual linear distortions, we evaluated the quality in terms of bit-error ratio (BER) and normalized generalized mutual information (NGMI) [16].

We used the following fiber parameters in the DBPs: a CD of 17.3 ps/nm/km, a dispersion slope of 0.062 ps/nm²/km, a nonlinear coefficient of 0.7 rad/W/km, a propagation loss of 0.192 dB/km, and a span length of 80 km. We considered DBPs with 1, 2, and 4-sps configurations in order to compare their performance-complexity trade-offs.

We optimized the reducing factor α and the bandwidth B of the LPF in DFF-DBP for each loop number at a fixed launch power of 3 dBm/channel. We performed a grid search over those parameters so that the Q factor converted from the BER, $Q_{\text{BER}} = \sqrt{2} \text{erfc}^{-1}(2 \times \text{BER})$, and the NGMI of the output signal were maximized simultaneously. Fig. 4 shows the optimized α and B for the 4-sps DFF-DBP as a function of loop number, and similar trends were observed for the 2- and 1-sps configurations. Both α and B decrease with increasing loop number; however, this trend does not imply performance degradation of DFF-DBP at longer transmission distances, as shown below.

We measured the BER of the signals for various launch powers and transmission distances after CDC and conventional or DFF-DBP schemes with 1-, 2-, and 4-sps configurations had been applied to the received waveforms. Fig. 5 shows the Q_{BER} as a function of launch power for loop numbers of 6 (988 km), 10 (1646 km), and 14 (2304 km). Similar Q_{BER} trends with respect to launch power are observed irrespective of transmission distance, and DFF-DBP consistently outperforms conventional DBP with the same sps configuration. In the 1-sps configuration, conventional DBP provides negligible performance improvement, whereas DFF-DBP

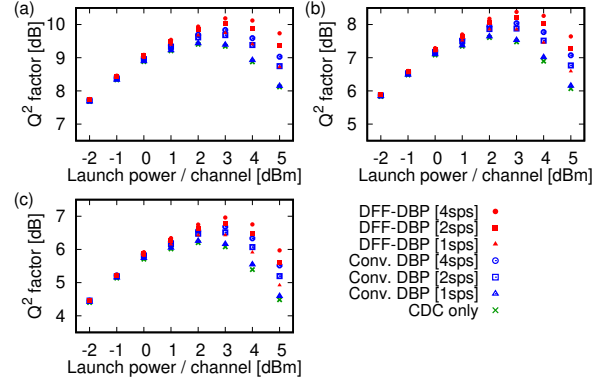


Fig. 5: Q_{BER} as a function of launch power for (a) 6 loops (988 km), (b) 10 loops (1640 km), and (c) 14 loops (2304 km).

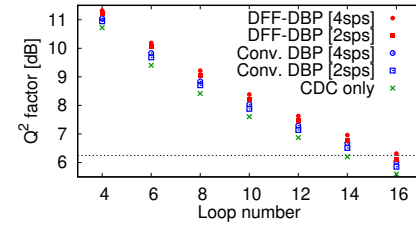


Fig. 6: Q_{BER} as a function of loop number. Dotted line is an SD-FEC threshold of 6.25 dB.

yields a noticeable gain at launch powers exceeding 1 dBm. In addition, the performance of 2-sps DFF-DBP surpasses that of 4-sps conventional DBP. For the 4-sps configuration, the Q_{BER}^2 gain of DFF-DBP over CDC was 0.78, 0.77, and 0.76 dB for 6, 10, and 14 loops, respectively. Thus, XPM-induced waveform distortion is effectively mitigated by DFF-DBP under all considered conditions.

Fig. 6 plots Q_{BER} as a function of the number of loops, where the launch power per channel was set to 2 dBm for the CDC-only case and to 3 dBm for all DBP schemes. At all transmission distances, DFF-DBP with 2- or 4-sps configurations provides superior signal quality compared with the other schemes. In particular, the Q_{BER}^2 gain of the 4-sps DFF-DBP over CDC only reaches 0.71 dB at 16 loops. Assuming an SD-FEC threshold of $Q_{\text{BER}}^2 = 6.25$ dB, corresponding to a BER of 2.0×10^{-2} , the maximum transmission distance is improved from 13 loops (2140 km) for CDC only to 16 loops (2634 km) for 4-sps DFF-DBP. As a result, DFF-DBP successfully extends the transmission distance by approximately 23 %.

Conclusions

We investigated performance improvement in long-distance WDM transmission systems using a 21-channel, 64-Gbaud DP-16QAM signal based on DFF-DBP. The design of DFF-DBP was improved to reduce its computational cost, and two key design parameters were optimized using experimentally obtained received waveforms. We then assessed the performance of the optimized DFF-DBP and confirmed that the achievable transmission distance was extended from 2140 km with CDC only to 2634 km with 4-sps DFF-DBP.

Acknowledgement

This work was partly supported by JSPS KAKENHI Grant Number JP26K07499.

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